

Project Executable Files

Date	15 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section documents all executable and deployable files for the OptiCrop project. All files are properly organized, named, and available on GitHub. The evaluator can access the live deployed application using the link below.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / dependency file	Yes
4	Database schema / seed files (if applicable)	NA — CSV dataset used
5	Environment configuration file (.env.example or similar)	NA — No API keys required
6	Deployed application URL (if hosted)	Yes — Deployed on Render
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA — Web application
8	Dockerfile / Containerization config (if applicable)	NA — Render handles deployment
9	Test files and test results	Yes — Performance Testing PDF
10	Demo video or walkthrough (if required)	Yes — Demo video uploaded to Drive

Step 2: File / Folder Structure

```
AI-ML-and-GEN-AI-Track-Project-Template/ ■■■ 1.Brainstorming & Ideation/ ■■■■ Brainstorming & Idea
Prioritization.pdf ■■■■ Define Problem Statements.pdf ■■■■ Empathy Map.pdf ■■■■ ER Diagram.pdf ■■■
2.Requirement Analysis/ ■■■■ Customer Journey Map.pdf ■■■■ Data Flow Diagram.pdf
■ ■■■■ Solution Requirements.pdf ■■■■ Technology Stack.pdf ■■■ 3.Project Design Phase/ ■■■■ Problem-Solution
Fit.pdf ■■■■ Proposed Solution.pdf ■■■■ Solution Architecture.pdf ■■■■ 4.Project Planning Phase/ ■■■■
Project Planning.pdf ■■■ 5.Project Development Phase/ ■■■■ OptiCrop.ipynb ■
■■■ app.py ■■■■ model.pkl ■■■■ Crop_recommendation.csv ■■■■ requirements.txt ■■■■ Procfile ■■■■
Coding & Solution.pdf ■■■■ templates/ ■■■■ home.html ■■■■ about.html ■■■■ findyourcrop.html ■■■
6.Project Testing/ ■■■■ Performance Testing.pdf ■■■■ 7.Project Documentation/ ■■■■ Project
Executable Files.pdf ■■■■ Sample Project Documentation.pdf ■■■■ 8.Project Demonstration/ ■■■■
Communication.pdf ■■■■ Demonstration of Proposed Features.pdf ■■■■ Project Demo Planning.pdf ■
■■■ Scalability & Future Plan.pdf ■■■■ Team Involvement in Demonstration.pdf ■■■ README.md
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://ai-ml-and-gen-ai-track-project-template-3.onrender.com
Login Credentials (Demo)	No authentication required — open access web application
Platform / Hosting Provider	Render (Free Tier) — https://render.com
Repository Link	https://github.com/BandaruBhanuprasad/OptiCrop-Smart-Agricultural-Production-Optimization-Engine
Demo Video Link	Google Drive link — to be added after recording demo video

Step 4: Run Instructions

Option 1 — Access Live Deployed App (Recommended):

Open browser and go to: <https://ai-ml-and-gen-ai-track-project-template-3.onrender.com>

Option 2 — Run Locally:

1. Clone the repository

```
git clone https://github.com/hemanth5267/AI-ML-and-GEN-AI-Track-Project-Template
```

2. Navigate to development folder

```
cd "AI-ML-and-GEN-AI-Track-Project-Template/5.Project Development Phase"
```

3. Create and activate virtual environment

```
python3 -m venv venv
```

```
source venv/bin/activate (Mac/Linux)
```

4. Install dependencies

```
pip install flask scikit-learn pandas numpy matplotlib seaborn jupyter
```

5. Run Jupyter Notebook to train and save model

jupyter notebook OptiCrop.ipynb → Run all cells → model.pkl will be generated

6. Run Flask application

```
python3 app.py
```

7. Open browser: <http://127.0.0.1:5000>

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Render free tier cold start — first request may be slow	Wait 30-60 seconds for first load — subsequent requests are fast
2	Logistic Regression convergence warning for low max_iter	Fixed by setting max_iter=1000 in model training
3	No user authentication in current version	Planned for future version — not required for current scope